NAUSICAA for ITACA

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Abstract

ITACA (Ion Tracking Apparatus with CMOS ASICs) is a proposed new detector to search for neutrinoless double beta decay ($\beta\beta0\nu$) events in a future, (multi) ton scale, ultra-low background, high pressure xenon TPC. The crucial innovation introduced by ITACA is the measurement of both the electron and the ion tracks produced in the xenon gas in response to ionizing particles. The readout of the ion track involves the use of electronics able to measure the charge carried by the positive xenon ions arriving at the cathode. We propose a design based on bare metal pads (TopMetal) coupled to CMOS electronics, which we call NAUSICA (Non-Amplified Ultra-Slow Ion CMOS ASIC).

This paper presents a conceptual design of NAUSICA and a simplified model of its response to ion tracks. We use a Geant4-based simulation of ITACA to generate $\beta\beta0\nu$ events uniformly in the fiducial volume of the detector. The ions produced by each track drift to the cathode where they are focused towards the metal electrodes. We calculate the distribution of ion arrival positions and times after drift, diffusion and focusing and the current induced on each collection pad. A charge-sensitive amplifier (CSA) computes the voltage produced as this current is integrated on a feedback capacitor with a finite discharge time. We then add electronic noise and digitise the voltage to obtain the digitised signal from each pad. The combined response of all the pads with signals above a noise-dependent threshold allows the reconstruction of the track. We show that the track can be resolved above the noise, keeping its characteristic topology.

1 Introduction

The next generation of neutrinoless double-beta decay ($\beta\beta0\nu$) experiments, aiming to establish if the neutrino is its own antiparticle, will attempt to explore lifetimes for a putative decay, $T_{1/2} > 10^{28}$ yr. This ambitious goal will require (multi) ton target masses of a double-beta decaying isotope and essentially zero backgrounds [1]. High Pressure Xenon TPCs (HPXe) [2] are one of the leading technologies for such a generation of large, virtually background-free $\beta\beta0\nu$ detectors. In particular, the NEXT collaboration is currently operating the NEXT-100 apparatus [3] at the Canfranc Underground Laboratory, a large scale detector which can hold ~ 100 kg of xenon at 15 bar, in a high pressure time projection chamber of roughly 1 m^3 volume. NEXT has already demonstrated a FWHM energy resolution of $(0.93 \pm 0.02)\%$ at $Q_{\beta\beta}$ [4].

An HPXe TPC provides a distinctive topological signature that enables identification of events with two emitted electrons. A $\beta\beta0\nu$ candidate is defined as a single, continuous track, with no

detached energy deposits and fully contained inside the fiducial volume. The energy of the track must be close to $Q_{\beta\beta}$ typically within a ROI of $\pm 1\sigma$. This "single-track" requirement suppresses nearly all backgrounds, leaving only single-electron events with energies near $Q_{\beta\beta}$, such as those arising from photoelectric interactions (without an accompanying X-ray satellite) of the 2.448 MeV γ ray of ^{214}Bi and the 2.615 MeV γ ray of ^{208}Tl, as well as from beta decays of ^{137}Xe created by cosmogenic activation. In addition, requiring large local energy deposits ("blobs") near the two endpoints of the track permits an additional suppression of single electron backgrounds that make it into the ROI. The effectiveness of this topological discrimination has been demonstrated by the NEXT collaboration [5–9].

However, the performance of the topological signature in an HPXeEL TPC is limited by diffusion and by electroluminescent blurring. As ionisation electrons drift through the dense gas, they undergo diffusion, producing a charge cloud whose transverse size increases with the drift length. In addition, EL amplification introduces further smearing, as the EL photons are emitted isotropically.

In two recent papers [10, 11], we proposed a new approach to improving the topological signature, while preserving EL amplification, needed for energy resolution. The method relies on recording two complementary tracks for each event: the *electron track*—the established signature in an HPXeEL TPC—and the *ion track*, formed by the positive ions left along the original particle trajectory. The ion track exhibits diffusion at the millimetre level and is free from EL-induced smearing, providing a dramatic boost to topological discrimination. The Ion Tracking Apparatus with CMOS ASICs (ITACA) [11] is a proposed high pressure xenon TPC based on this principle. The ions are read out using bare metal pads (TopMetal) coupled to CMOS electronics.

In this paper we present a conceptual design of NAUSICA (Non-Amplified Ultra-Slow Ion CMOS ASIC), our proposed readout for ITACA. Our study [11] showed that the topological signature that allows distinguishing between signal ($\beta\beta0\nu$ events which produce two electrons) and single-electron backgrounds could increase the rejection power of the HPXe technology by at least one order of magnitude, provided that the ion track could be reconstructed with a spatial resolution at the level of ion diffusion ($\sigma \sim$ 1–2 mm). Here we show that the proposed NAUSICA readout can preserve the resolution at the level dictated by the diffusion, thus guaranteeing the expected enhancement of the topology discrimination power.

This paper is organized as follows. Section 2 briefly summarizes the main aspects of the ITACA detector. Section 3 presents the NAUSICA concept.

2 The ITACA Detector

The pressure of operation in ITACA is chosen as a tradeoff between mass — which increases with pressure — and track topology, which is easier to resolve at lower pressure. Other considerations include ion drift velocity, which decreases approximately inversely with pressure at fixed electric field, and electron lifetime, which tends to improve at lower pressure. Overall, 15 bar is a good compromise. Using electroluminescence to amplify the electron track signal guarantees excellent energy resolution. Fiducialisation of the event requires the measurement of t_0, provided by primary scintillation. Both requirements are essential for a $\beta\beta0\nu$ experiment.

Choosing the detector dimensions implies a tradeoff between the diameter and the length of the TPC. A major design goal is to minimise the transverse diffusion σ_T of ions, which in the thermal limit is simply:

$$\sigma_T = \sqrt{\frac{2k_B T}{q_e}\frac{L}{E}} \tag{1}$$

where L is the ion drift distance, E the drift field, T the gas temperature, k_B Boltzmann's constant and q_e the elementary charge. Equation 1 favours shorter L and larger E. Figure 1 shows σ_T for three different drift fields. For a TPC drift length of 150 cm, the average σ_T is of the order

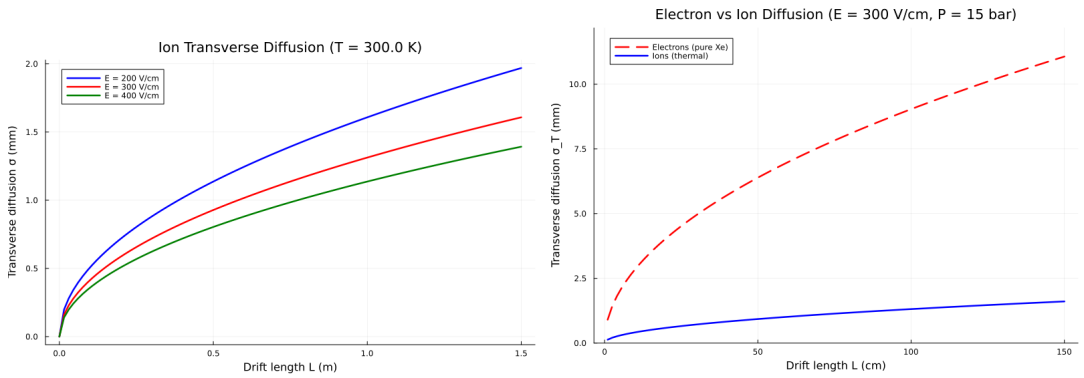


Figure 1: Left: transverse diffusion of Xe_2^+ ions as a function of drift length for three different drift fields; right: transverse diffusion of Xe_2^+ ions compared with transverse diffusion of electrons in pure xenon.

Table 1: Summary of ITACA detector parameters.

Parameter	**Symbol**	**Value**	**Unit**
Geometry			
Fiducial diameter	D_{fid}	320	cm
Fiducial height	L_{fid}	150	cm
Fiducial volume	V_{fid}	12.1	m^3
Fiducial Xe mass		1050	kg
PV internal diameter		3.54	m
PV internal height		2.01	m
Electric field and drift			
Drift field	E	200	V/cm
Electron drift velocity	v_e	1	mm/μs
Electron drift time	t_e	1.5	ms
Ion drift velocity	v_d	10	cm/s
Ion drift time (max)	t_{max}	15	s

of 1 mm, one order of magnitude smaller than the corresponding diffusion for electrons in pure xenon.

Another important design factor is the voltages in the TPC electrodes. All sensors (the CMOS reading the ions and the SiPMs reading the EL light) operate best at ground, but the drifting of electrons and ions requires a constant field in the TPC, and the generation of EL light requires an additional, higher field, in the EL region.

The cathode is instrumented with an Ion Focusing Grid (IFG) which focuses the incoming ions into the readout pads of the CMOS detector. The upper electrode of the IFG is at a voltage $V_c = 8\,\mathrm{kV}$. The voltage on the anode side of the TPC depends on the drift length and the drift field. A good compromise between minimising diffusion and keeping voltage at the anode reasonable is to choose a drift field of 200 V/cm, which results in 38 kV at the first anode electrode, and 54 kV at the second (with 16 kV across the EL gap).

Once the length of the TPC is defined, the diameter of the TPC defines the fiducial mass that can be achieved. At 15 bar, choosing $D = 320$ cm, results in a fiducial mass slightly above one ton, $M_{fid} = 1050$ kg.

A large diameter TPC requires instrumenting both the EL side (with SiPMs) and the cathode side (with top metal devices). Previous work involving the detection of ions for $\beta\beta0\nu$ experiments [12, 13] has quantified the challenge of instrumenting a large surface with CMOS detectors, which appears, nonetheless, feasible [14]. The ITACA paper [11] proposes a movable ion detector, the Magnetically Actuated Rotor System (MARS). The NAUSICA readout considered here could be used as part of MARS or to instrument the full cathode. Table 1 shows the main parameters of a 1-ton ITACA detector.

2.1 Principle of operation

Consider a putative $\beta\beta0\nu$ event occurring in the middle of the TPC. Two electrons, with a total energy of $Q_{\beta\beta}$ (2458 keV), propagate through the dense noble gas, ionising it. Primary scintillation is recorded and a fast trigger system (FTS) records the start-of-the-event t_0. Then a fast measurement of the event energy is performed, accepting the event only if its energy lies within a pre-determined ROI (selected to be $\pm1\sigma$ around $Q_{\beta\beta}$). Events outside the ROI are rejected and no ion readout is attempted.

Since the event that we are considering is a $\beta\beta0\nu$ decay, the FTS will likely trigger, activating a fast reconstruction of the topology and event barycenter, (x_i, y_i, z_i). The coordinates (x_i, y_i) predict the landing position of the ion cloud at the cathode, while z_i, together with t_0 and the ion drift velocity, determines the arrival time of its barycenter (t_i).

The Xe_2^+ ions reach the cathode on a time scale of seconds. In our specific example, with the event at the midpoint of the TPC, they need to drift 75 cm at a velocity of 10 cm/s, requiring 7.5 s. The typical longitudinal extension of the ion track is of the order of 10 cm, and thus about 1 s is needed to collect the ions.

Notice that the electron track and the ion track are detected in *delayed coincidence*. The ion track is intercepted precisely at the right time and in the right place, thus minimising the impact of spurious coincidences. At the expected trigger rate (dominated by backgrounds in the ROI), pile-up during the ∼15 s maximum ion drift time is negligible.

3 The NAUSICA concept

Reading the arriving ions (without amplification) requires exposed metal electrodes connected to charge-sensitive amplifiers (CSAs). Our proposed *Non-Amplified Ultra-Slow Ion CMOS ASIC*

(NAUSICA) uses TopMetal CMOS technology [15] to implement this concept, following Refs. [12, 13]. There is an important difference, however, with respect to these works, which attempted to design a system able to provide simultaneously good energy resolution and a good measurement of the track topology using ions. Alas, both requirements impose constraints that interfere with each other. Good energy resolution requires optimizing ion counting and minimizing noise. Good topological reconstruction demands pixels separated by a small pitch (e.g., of the order of diffusion, ~ 2 mm), which in turn multiplies the number of pixels and adds to the overall noise. The advantage of ITACA is that energy resolution is measured with excellent precision by electroluminescence, and thus the ion detector can prioritize the reconstruction of the topology. NAUSICA is designed with this goal in mind. Appendix D compares its geometry and charge-injection noise with Topmetal-S.

Consider the $\beta\beta0\nu$ event that we described in the previous section. Assume the extreme case in which the track extends along z. For a track length of 10 cm and a drift velocity of 10 cm/s it would take one second, since the arrival of the first ions, to collect the full track. In practice, the tracks are oriented in random directions, extending radially over several pads, and collection of ions is faster, since few tracks extend exactly along z. In any case, the resolution in z is of the order of the transverse diffusion, ($\sigma_z = 1 - 2$ mm), so reading the charge in frames of 10-20 ms is reasonable. We take 20 ms as the nominal frame duration, corresponding to 2 mm along the drift direction.

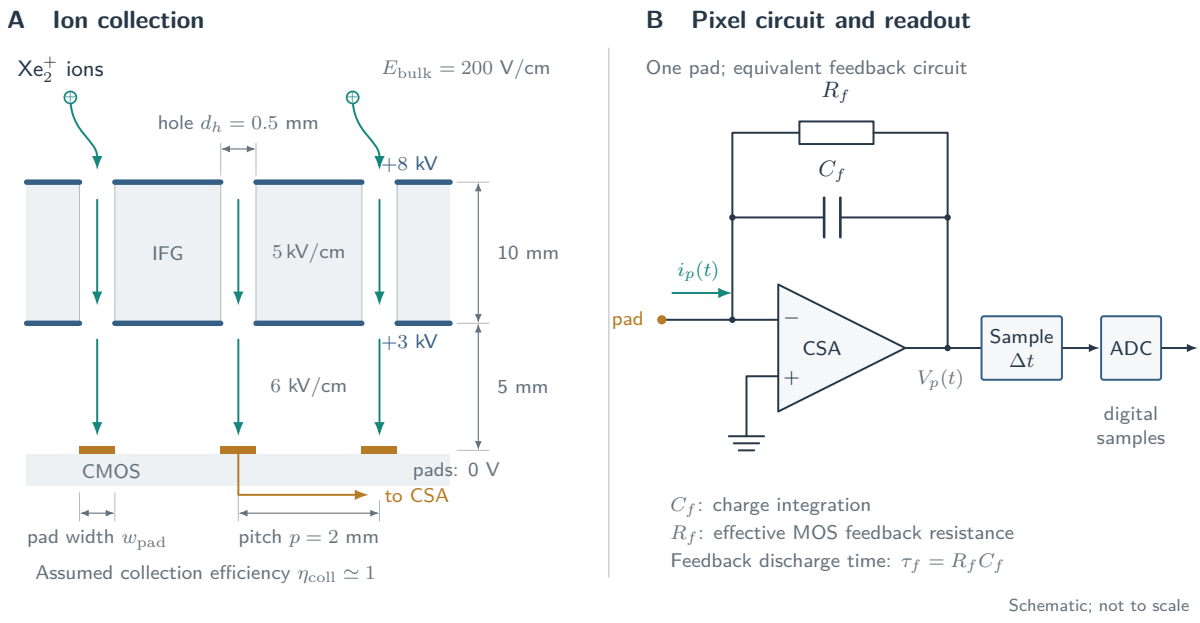


Figure 2: NAUSICA ion collection geometry (left) and equivalent pixel circuit (right). The drawing is schematic and not to scale.

3.1 Charge collection and focusing

The pad dimensions and pitch need to be chosen balancing the desired spatial resolution with the need for an acceptable signal-to-noise (S/N) per pad. Larger pads increase the collection but also the input capacitance, thus the need to focus the incoming ions in an area as small as possible. The left panel of Fig. 2 shows a schematic of the Ion Focusing Grid (IFG), which does this job. We estimate the focusing requirement by comparing the electric flux through one pixel cell with that through its hole. Taking uniform fields in the drift region and inside the hole, full

flux collection requires

$$E_h \frac{\pi d_h^2}{4} \gtrsim E_{\text{bulk}} p^2.$$

Here E_h and E_{bulk} are the fields inside the hole and in the drift region, respectively, d_h is the hole diameter, and p is the pixel pitch. For a pitch of 2 mm, a bulk field of 200 V/cm and a hole field of 5 kV/cm, this gives a minimum hole diameter of approximately 0.45 mm. Thus, we take 0.5 mm as a preliminary design parameter. We use square pads of 0.5 mm side on a 2 mm pitch. Full flux collection in this simple geometry implies a large ion collection efficiency that we take as one for the time being.

3.2 Pixel electronics and readout

The right panel of Fig. 2 shows a schematic of the electronics. Each pad is directly connected to the input of a CSA. The input current is integrated on a feedback capacitor C_f. A MOS transistor allows C_f to discharge slowly. For an ideal CSA, the gain is $1/C_f$ and the signal decays with time constant τ_f.

The sampled voltage includes charge accumulated in previous buckets. The charge arriving in each bucket is recovered from the change between successive samples, accounting for feedback discharge. We use frames of 20 ms. Each voltage reading is averaged over 300 µs, as described in Section 5.3. This requires repeated digitisation within the acquisition window or an averaging stage before the ADC.

Digitisation could use ADCs shared by groups of pixels and mounted on readout boards behind the collection plane. Buffered pixel outputs would be multiplexed to these ADCs, with sufficient sampling to perform the averaging described above.

3.3 Parameters of the design

In summary, the pad size, pitch and active area must be chosen to sample and contain the ion track. The feedback capacitance and discharge time depend on the charge arriving at each pad and its distribution in time. The sampling interval, electronic noise and ADC range must allow the weaker parts of the track to be recorded while accommodating the largest signals.

4 Signal formation and response of one pad

Consider an ion, with charge $q = e$, arriving at a readout pad. The process of signal formation can be viewed in two steps:

1. As the ion approaches, a current is induced in the pad.
2. Then the CSA integrates it and produces the voltage sampled by the readout.

Next, we describe each step:

4.1 The induced current

The weighting potential of a pad, $\phi_w(r)$, describes how an ion's motion contributes to the signal on that pad. We calculate it for the detector geometry with the following boundary conditions: (a) the selected pad is held at unit potential and (b) all other electrodes at zero. Also the calculation is performed in the absence of charge.

We define positive current as current delivered into the CSA input by an approaching positive ion. With this convention, the meaning of ϕ_w follows from the Shockley–Ramo relation [16]

$$\Delta Q = e\,\Delta\phi_w. \tag{2}$$

The current is the rate at which this signal develops:

$$i_1(t) = e\frac{\mathrm{d}\phi_w(\mathbf{r}(t))}{\mathrm{d}t}. \tag{3}$$

Near a small pad, ϕ_w changes rapidly with position, so most of the signal is induced during the final approach.

For the analytical calculation we use a square pad of side $2a$, embedded in an infinite grounded plane. The gas occupies a homogeneous half-space above the plane. The ion approaches the centre of the pad along its normal at constant speed v. The gap width g is the distance between the lower IFG electrode and the pad plane. This geometry provides a simple pad response. In this simple approximation, the IFG and the dielectric structure of the CMOS surface are not included in its weighting potential. Figure 3 illustrates the geometry and weighting-potential boundary conditions. Appendix A gives the derivation.

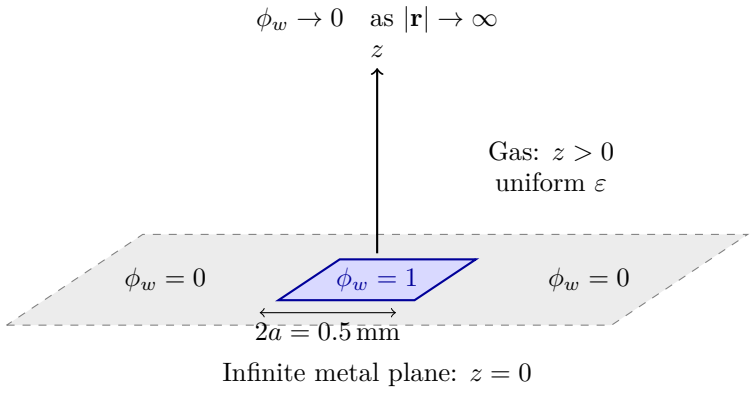


Figure 3: Boundary conditions for the square-pad weighting potential. The drawing shows a portion of the infinite metal plane; the dashed outline indicates its continuation. The pad has $\phi_w = 1$ and the surrounding metal $\phi_w = 0$. In the gas, the potential satisfies Laplace's equation and tends to zero at infinity. Schematic, not to scale.

Place the pad at $z = 0$ and take collection to occur at $t = 0$, so that $z = -vt$ before collection. The weighting field on the central axis is

$$W(z) = -\frac{\mathrm{d}\phi_w}{\mathrm{d}z} = \frac{4a^2}{\pi(z^2 + a^2)\sqrt{z^2 + 2a^2}}. \tag{4}$$

For an ion entering this gap at $z = g$ and $t_g = -g/v$, the current is

$$i_1(t) = \begin{cases} evW(-vt), & t_g \le t < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Its integral over this finite trajectory is $Q_g = e[1 - \phi_w(g)]$.

4.2 Signal size and duration

As a numerical example, we take the parameters shown in the left panel of Fig. 2: a pad of side 0.5 mm and a gap of 5 mm. We estimate the gap velocity from $v = \mu E_g$, using the mobility inferred

from the bulk drift speed and assuming it remains constant in the gap. With $E_g = 6\,\mathrm{kV/cm}$, this gives $v = 3\,\mathrm{m/s}$.

The current rises as the ion approaches the pad and reaches

$$i_1(0^-) = \frac{2\sqrt{2}}{\pi}\frac{ev}{a} = 1.731\,\mathrm{fA}. \tag{6}$$

Most of the signal is induced close to the pad (Figure 4). The final ∼200 µs before collection contain 90% of the full-path charge e [1]. For the finite gap used in the calculation (Appendix A.2),

$$Q_g = e[1 - \phi_w(g)] = 0.9984124\,e. \tag{7}$$

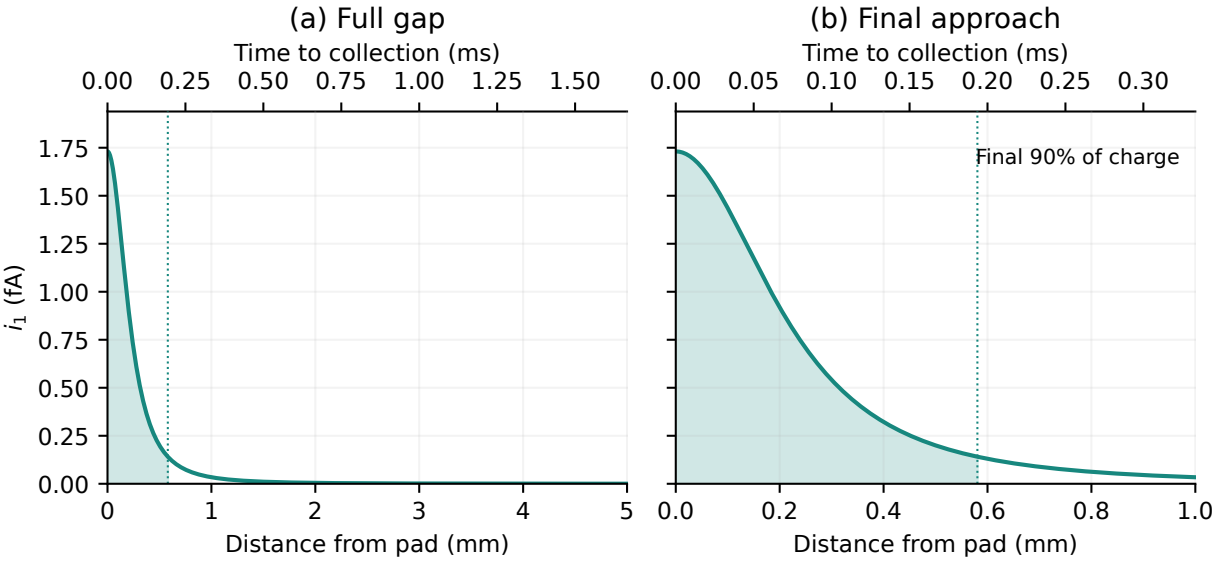


Figure 4: Single-ion current across the full gap (left) and during the final approach (right), for the square-pad model described above. The upper axes show the time remaining before collection. The shaded region contributes 90% of the full-path charge.

4.3 Ideal CSA response

We first describe the CSA by an ideal amplifier with feedback capacitance C_f and an effective linear resistance R_f representing the MOS discharge path. Consider an ion approaching the pad and reaching it at $t = 0$. The ion induces an input current $i_1(t)$, shared between the feedback capacitor and the discharge path. The resulting output voltage V_1, measured relative to its baseline, satisfies:

$$i_1(t) = -C_f\frac{\mathrm{d}V_1}{\mathrm{d}t} - \frac{V_1}{R_f}. \tag{8}$$

The first term describes charge integration and the second feedback discharge. With the discharge time $\tau_f = R_f C_f$, this becomes

$$\frac{\mathrm{d}V_1}{\mathrm{d}t} + \frac{V_1}{\tau_f} = -\frac{i_1(t)}{C_f}. \tag{9}$$

We assume that no signal is stored in the CSA when the ion enters the gap at $t = t_g$. Since V_1 is measured relative to the baseline, $V_1(t_g) = 0$, and the solution is

$$V_1(t) = -\frac{1}{C_f}\int_{t_g}^{t} i_1(t')\exp\left[-\frac{t-t'}{\tau_f}\right]\mathrm{d}t'. \tag{10}$$

[1] Here "full path" means extending the same analytical trajectory to infinite distance.

With this convention positive input current produces a negative voltage. Once collection is complete, the voltage decays exponentially, $V_1(t) = V_1(0)\exp(-t/\tau_f)$.

Neglecting feedback discharge during the approach of one ion, the integral at collection, $t = 0$, gives

$$V_1(0) = -\frac{1}{C_f}\int_{t_g}^{0} i_1(t')\,\mathrm{d}t' = -\frac{Q_g}{C_f}.$$

For $C_f = 3\,\mathrm{fF}$, its magnitude is $53.32\,\mu\mathrm{V}$.

For several ions following the same final trajectory and collected on the same pad at times t_k, the currents and voltages add:

$$i_p(t) = \sum_k i_1(t - t_k), \qquad V_p(t) = \sum_k V_1(t - t_k). \tag{11}$$

The voltage therefore includes contributions from earlier arrivals that have not yet discharged.

5 CSA response and noise

A real amplifier has finite gain and bandwidth. The output voltage for a given charge is smaller than the ideal value Q/C_f, and takes time to develop. Electronic noise also sets a limit on the minimum charge that can be measured. We calculate these effects using a transistor circuit in SPICE, with the ion-induced current from Section 4 as its input.

5.1 Circuit and operating parameters

Figure 5 shows the CSA used in the calculation. It has a PMOS input transistor, a folded-cascode amplifier and a feedback capacitor C_f. An NMOS transistor, M_f, provides the discharge path. The circuit uses 180 nm CMOS device models and a supply of 1.8 V. The transistor dimensions are given in Table 2. The DC operating-point calculation gives an output voltage of 0.879 V and a quiescent power consumption of $344\,\mu\mathrm{W}$ for the CSA.

Device	Type	W (μm)	L (μm)	m
M1	PMOS	220	0.5	1
M2	NMOS	27	0.5	3
M3	NMOS	200	2	1
M4	NMOS	2.88	1.44	1
M5	PMOS	220	0.5	1
M6	PMOS	1	6	1
M_f	NMOS	0.36	100	1

Table 2: Transistor dimensions used in the SPICE calculation.

The pad capacitance $C_{\mathrm{pad}} = 1.25\,\mathrm{pF}$ for our 0.5 mm pads is estimated from conservative scaling of the Topmetal-S capacitance [17]. Table 3 lists the circuit and readout parameters used below.

5.2 Response to uniform arrivals

We consider N ions arriving uniformly over a frame of duration Δt. The number of ions is left to be determined from the required S/N. Averaging the single-ion current over the collection time u gives

$$i_N(t) = \frac{N}{\Delta t}\int_0^{\Delta t} i_1(t - u)\,\mathrm{d}u. \tag{12}$$

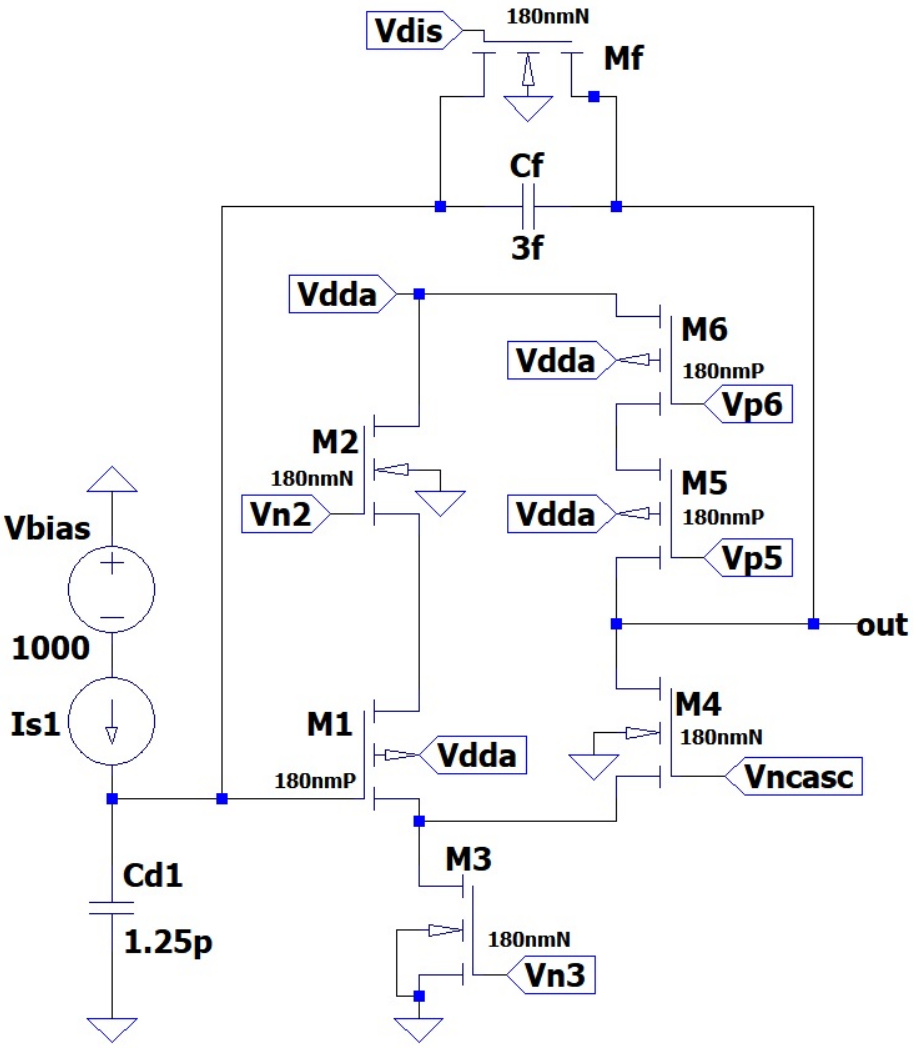


Figure 5: CSA circuit used in the SPICE calculation. The pad current enters the gate of M1; C_f and M_f connect the output to this input node. Transistor dimensions and operating parameters are given in Tables 2 and 3.

Parameter	Symbol	Value
External pad capacitance	C_{pad}	1.25 pF
Feedback capacitance	C_f	3 fF
Supply voltage	V_{DD}	1.8 V
M3 gate voltage	V_{nload}	0.5385 V
M4 gate voltage	V_{ncasc}	0.5568 V
Temperature	T	293.15 K
Frame duration	Δt	20 ms
Averaging time	T_a	300 µs
Discharge-correction time	τ_0	2.454 s
Noise integration band	$[f_\ell, f_h]$	0.1 Hz–1 MHz

Table 3: Parameters of the circuit and frame measurement. Gate bias voltages are rounded to four decimal places. The time τ_0 is obtained from the simulated voltage decay and used in the correction between readings.

The currents overlap, producing an approximately constant current through most of the frame. The voltage builds up during this interval. The current is supplied to SPICE as a table of times and values, with its integral fixed by the finite-gap charge NQ_g.

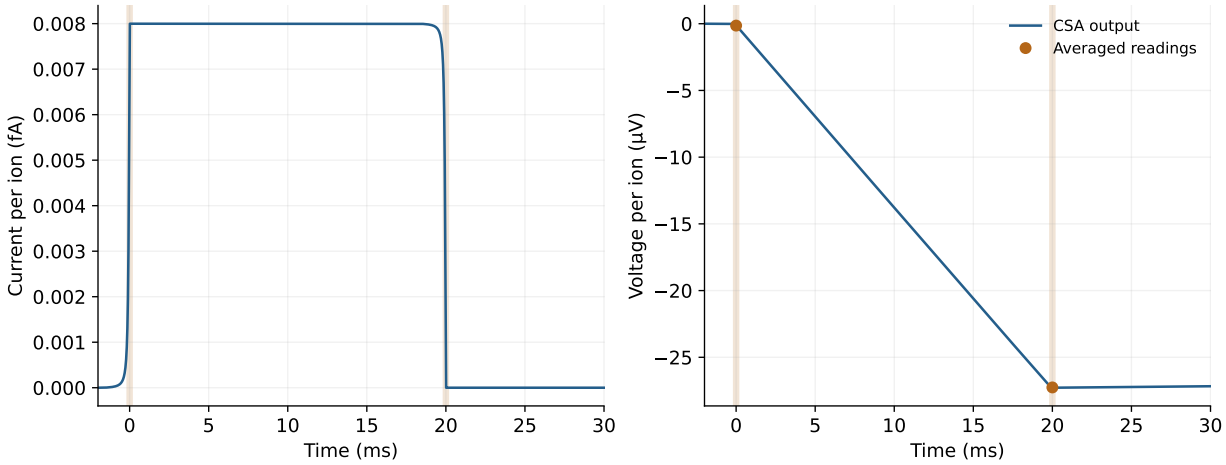


Figure 6: Uniform ion arrivals over a 20 ms frame. Left: input current per ion. Right: SPICE output voltage relative to the baseline, per ion. The shaded windows have width 300 µs; the points show the averaged voltages.

Figure 6 shows the current and voltage normalized to one ion. Since the voltage remains after the arrival interval, successive frames must be treated as a continuous signal.

5.3 Frame readout

The voltage at each frame boundary is averaged over a window of duration $T_a = 300\,\mu\text{s}$, large enough to average the rapid voltage fluctuations that would enter a point reading, but much shorter than the 20 ms frame.

$$\overline{V}_n = \frac{1}{T_a}\int_{t_n - T_a/2}^{t_n + T_a/2} V(t)\,\mathrm{d}t, \qquad t_n - t_{n-1} = \Delta t. \tag{13}$$

When discussing the ideal CSA, we saw that the residual voltage decays exponentially, $\exp(-\Delta t/\tau_f)$. This motivates the corrected voltage difference

$$D_n = -\left(\overline{V}_n - \alpha \overline{V}_{n-1}\right), \qquad \alpha = \exp(-\Delta t/\tau_0). \tag{14}$$

which takes into account the voltage left from the previous frame. The minus sign makes D_n positive for the chosen output polarity. We determine the value of τ_0 ($\tau_0 = 2.454\,\text{s}$) by fitting an exponential to the SPICE output voltage, measured relative to its baseline, after the ion current has ended. We then calibrate the resulting measurement with the SPICE response. Let g be D_n for the uniform current normalized to one ion. In the linear regime, N ions give Ng, and their estimated number is

$$\widehat{N}_n = \frac{D_n}{g}. \tag{15}$$

Applying the averaging and discharge correction to the simulated response gives $g = 27.119\,\mu\text{V}$ per ion.

For equal-sized ion packets, each neighbouring frame contributes less than 1% of the signal in the selected frame.

5.4 Noise and sensitivity

The amplifier and the feedback transistor produce voltage fluctuations even in the absence of ions. The transistor models include thermal (white) noise and flicker noise —which increases approximately as $1/f$ at low frequencies—. SPICE calculates their contributions to the output noise and sums their power spectral densities. We denote the resulting one-sided output spectrum by $S_V(f)$, in V^2/Hz.

The noise in the measured difference is

$$\delta D_n = -\left(\delta\overline{V}_n - \alpha\delta\overline{V}_{n-1}\right). \tag{16}$$

Thus the uncertainty depends on the noise in both readings. Positively correlated fluctuations partly cancel when we take their difference. Averaging the voltage over the acquisition window and taking the corrected difference between successive readings gives

$$\sigma_D^2 = \int_{f_\ell}^{f_h} S_V(f) \left|\frac{\sin(\pi f T_a)}{\pi f T_a}\right|^2 \left|1 - \alpha e^{-2\pi \mathrm{i} f \Delta t}\right|^2 \mathrm{d}f. \tag{17}$$

The factor $|\sin(\pi f T_a)/(\pi f T_a)|^2$ accounts for averaging over the acquisition window T_a. The factor $\left|1 - \alpha e^{-2\pi \mathrm{i} f \Delta t}\right|^2$ describes the corrected difference between readings separated by Δt. Appendix B.3 derives this expression from the covariance between readings.

Integrating from 0.1 Hz to 1 MHz gives $\sigma_D = 3.056\,\mathrm{mV}$. Dividing by the signal per ion gives the input-equivalent noise and the signal-to-noise ratio:

$$\mathrm{ENC} = \frac{\sigma_D}{g} = 112.7 \text{ ions RMS}, \qquad \mathrm{S/N}(N) = \frac{Ng}{\sigma_D} = \frac{N}{\mathrm{ENC}}. \tag{18}$$

Each ion carries charge $+e$. Combining the simulated voltage response with the calculated noise gives $\mathrm{S/N} > 5$ for $N \geq 564$ ions.

Table 4 gives the transistor contributions to the noise variance after averaging and taking the corrected frame difference.

Transistor	Contribution (%)
M_f	48.9
M_4	25.2
M_2	18
M_3	4.5
M_1, M_5, M_6	3.5

Table 4: Contributions to the noise variance σ_D^2 for the selected circuit and readout parameters.

The dependence with the averaging windows of the ENC is very soft. It varies from 116.4 to 111.7 electrons RMS for windows between 180 µs and 350 µs.

5.5 Pixel area and integration

The 2 mm pitch provides $4\,\mathrm{mm}^2$ per pixel. We estimate that the pad, CSA and space reserved for readout and control occupy approximately $1\,\mathrm{mm}^2$. Figure 7 shows their arrangement within the pixel cell, and Table 5 gives the area budget. The CSA dimensions follow from the transistor sizes in Table 2, with space around each device for contacts and connections. The remaining electronics are represented by area reserves. The shared ADCs on the rear readout boards are excluded from this budget. Appendix C describes the packing assumptions.

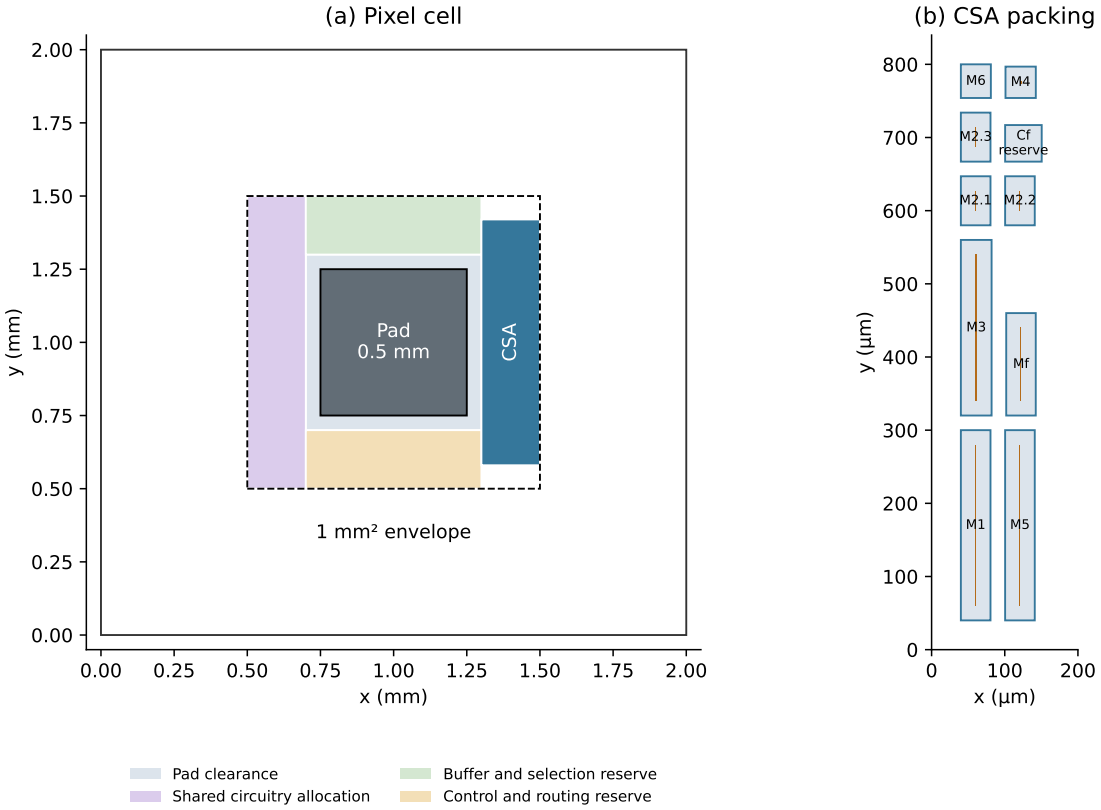


Figure 7: Pixel area allocation. Left: the pad, CSA and reserves for readout, control and shared circuitry inside a 1 mm^2 envelope, shown within the full pixel cell. The shared circuitry block represents its area allocation per pixel. Right: packing of the transistor channels and their surrounding space, together with a reserve for the feedback capacitor.

Block	Area (mm^2)
Pad and clearance	0.36
CSA and feedback	0.17
Buffer and selection reserve	0.12
Control and routing reserve	0.12
Shared circuitry allocation	0.20
Total	0.97

Table 5: Pixel area budget for the arrangement in Fig. 7.

A Analytical single-ion response

A.1 Weighting potential of a square pad

The pad occupies $-a < x < a$, $-a < y < a$ in the plane $z = 0$. Its weighting potential is one, and the remainder of the plane is held at zero. The upper half-space has uniform permittivity and no free charge in the weighting problem. The potential tends to zero at infinity. Thus $\nabla^2\phi_w = 0$ for $z > 0$, with a prescribed potential on the boundary.

Figure 3 illustrates these boundary conditions:

$$\phi_w(x, y, 0) = \begin{cases} 1, & |x| < a,\ |y| < a \quad \text{(pad)}, \\ 0, & \text{elsewhere on the plane}, \end{cases} \qquad \phi_w(\mathbf{r}) \longrightarrow 0 \quad \text{as } |\mathbf{r}| \longrightarrow \infty. \tag{19}$$

The values one and zero define the dimensionless weighting potential for this pad. The operating voltages determine the physical field that drives the ions.

The half-space Poisson kernel gives [18]

$$\phi_w(x, y, z) = \frac{1}{2\pi} \int_{-a}^{a} \int_{-a}^{a} \frac{z\,\mathrm{d}x'\,\mathrm{d}y'}{[(x - x')^2 + (y - y')^2 + z^2]^{3/2}}. \tag{20}$$

The integral is the solid angle subtended by the pad, divided by 2π. On the central axis, integration over the square yields

$$\phi_w(z) = \frac{2}{\pi} \tan^{-1}\left[\frac{a^2}{z\sqrt{z^2 + 2a^2}}\right], \qquad z > 0. \tag{21}$$

It approaches one at the pad and zero far from it. Differentiation gives the field in Eq. (4). At large distance,

$$\phi_w(z) \simeq \frac{2a^2}{\pi z^2}, \qquad W(z) \simeq \frac{4a^2}{\pi z^3}. \tag{22}$$

The weighting field has units of inverse length because ϕ_w is dimensionless.

This boundary condition represents a metal pad surrounded by grounded metal. An exposed dielectric surface between pads would require a different solution. The actual IFG also modifies the weighting potential. The present expression is used as an analytical reference, with fixed electrode potentials and negligible perturbation of the physical field by the ion charge. Resistive materials can require time-dependent weighting fields [16, 19].

A.2 Trajectory, charge and duration

The bulk operating point gives the mobility

$$\mu = \frac{v_d}{E_{\text{bulk}}} = 5 \times 10^{-6}\ \mathrm{m}^2/(\mathrm{V\,s}). \tag{23}$$

We use the same mobility in the nominal uniform gap field, giving $v = \mu E_g$. Diffusion during the final approach is omitted from the single-ion reference trajectory. The ion moves from $z = g$ to zero at this constant speed. Its induced current is Eq. (5), and direct integration gives

$$\int_{-g/v}^{0} i_1(t)\mathrm{d}t = e \int_0^g W(z)\mathrm{d}z = e[1 - \phi_w(g)]. \tag{24}$$

For the example in Section 4, $\phi_w(g) = 0.0015876$. The finite-path signal is not rescaled to e. Extending this trajectory to infinite distance gives $\int i_1 \mathrm{d}t = e$, but does not model passage through the IFG.

Inverting Eq. (21) gives

$$z(\phi) = a\sqrt{\frac{1}{\sin(\pi\phi/2)} - 1}. \tag{25}$$

For the full reference trajectory, the interval immediately preceding collection that contains a fraction F of the charge has duration

$$T_F = \frac{z(1-F)}{v}. \tag{26}$$

For a finite starting height g, the corresponding duration is

$$T_{F,g} = \frac{z(1-F[1-\phi_w(g)])}{v}. \tag{27}$$

These definitions refer to integrated charge, not to the time above a fraction of the peak current. The pulse amplitude scales as ev/a, and its characteristic duration as a/v. Their product preserves the full-path charge.

For comparison, a circular pad of radius a in the same grounded plane has

$$\phi_{w,\circ}(z) = 1 - \frac{z}{\sqrt{z^2+a^2}}, \qquad W_\circ(z) = \frac{a^2}{(z^2+a^2)^{3/2}}. \tag{28}$$

Its central peak is $ev/a = 1.923\,\mathrm{fA}$ for the same lateral diameter as the square side, compared with 1.731 fA for the square. Both results assume the same constant velocity and grounded surroundings. Their difference illustrates shape dependence; it does not quantify the uncertainty of the real IFG–pad structure.

A.3 Integration and feedback discharge

For an ideal CSA, the input node remains at its reference potential. Current balance through the parallel feedback elements gives

$$C_f \frac{\mathrm{d}V_1}{\mathrm{d}t} + \frac{V_1}{R_f} = -i_1(t). \tag{29}$$

Multiplication by $\exp(t/\tau_f)$ and integration from $t_g = -g/v$ gives Eq. (10). In the absence of discharge,

$$V_1(t) = -\frac{e}{C_f}[\phi_w(-vt) - \phi_w(g)], \qquad t_g \le t \le 0. \tag{30}$$

The voltage remains at $-Q_g/C_f$ after collection. With finite τ_f, its value at collection can also be written as

$$V_1(0) = -\frac{e}{C_f}\int_0^g W(z)\exp\left(-\frac{z}{v\tau_f}\right)\mathrm{d}z. \tag{31}$$

This form permits an independent numerical check of the time-domain calculation.

Integrating the CSA equation between two readout times t_{n-1} and t_n gives the exact charge relation

$$Q_n = -C_f\left[V_n - V_{n-1} + \frac{1}{\tau_f}\int_{t_{n-1}}^{t_n} V(t)\mathrm{d}t\right], \tag{32}$$

where Q_n is the induced input charge in that interval and $V_n = V(t_n)$. Thus the difference between samples gives the charge when discharge is negligible. With finite discharge, the correction depends on the voltage between samples. Recovering it from discrete samples requires a model or approximation for that interval.

If the current is constant over an interval Δt, its charge and the sampled voltages satisfy

$$Q_n = -C_f \frac{\Delta t/\tau_f}{1-\exp(-\Delta t/\tau_f)} \left[V_n - \exp(-\Delta t/\tau_f) V_{n-1}\right]. \tag{33}$$

The limit $\tau_f \to \infty$ is $Q_n = -C_f(V_n - V_{n-1})$. Equation (33) assumes constant current within the interval; two endpoint samples do not otherwise determine an arbitrary arrival history.

A.4 Numerical evaluation and checks

The current is represented by charge increments rather than point samples. For a time bin $[t_j, t_{j+1}]$ during the approach,

$$q_j = e[\phi_w(-vt_{j+1}) - \phi_w(-vt_j)]. \tag{34}$$

These increments sum exactly to the finite-path charge. Taking the bin-average current $q_j/\delta t$ gives the recurrence

$$V_{j+1} = \alpha V_j - \frac{q_j}{C_f}\frac{\tau_f}{\delta t}(1-\alpha), \qquad \alpha = \exp(-\delta t/\tau_f). \tag{35}$$

This is exact for a constant current in each numerical bin and converges to the continuous-current solution as δt decreases. The numerical step is separate from the readout sampling interval.

The square-pad potential is checked against independent surface integration of Eq. (20), and its derivative against Eq. (4). Charge integrals, boundary limits and scaling with pad size and velocity are also checked. The CSA response is compared with Eq. (31), the ideal-integrator limit and the closed-form response to a constant current. These checks test the analytical model and its numerical implementation; they do not replace a field calculation for the collection geometry or a transistor-level circuit calculation.

B Circuit and frame measurement

B.1 Circuit parameters

The calculation uses BSIM3 transistor models for 180 nm CMOS, evaluated with LTspice 17.2.4. Figure 5 gives the connections and gate biases. Table 2 gives the dimensions, including the multiplicity m of parallel devices. The pad capacitance is an external load; intrinsic device capacitances are supplied by the transistor models.

The gate voltages of M3 and M4 are quoted as 0.5385 V and 0.5568 V, respectively. Repeating the calculation with these rounded values changes the gain by 0.27% and the ENC by 0.06%.

The transient voltage is measured relative to a matched zero-input run. The physical current has the square-pad boundary conditions specified in Appendix A; its integral is preserved in the tabulated SPICE source. The gain g is evaluated with the uniform arrival current, the averaging windows and the frame difference defined in the main text.

B.2 Contributions from neighbouring frames

For equal-sized ion packets, the voltage left by a packet in the preceding frame contributes 0.082% of the selected-frame signal after the discharge correction. A packet collected in the following frame contributes 0.518%. Its ions induce current before reaching the pad, and the voltage averaging window extends across the frame boundary. Both contributions are calculated with the same averaging and discharge correction used to recover the charge.

B.3 Averaging and noise covariance

Let $C_{\overline{V}}(u)$ be the covariance of averaged voltage readings separated by u. For stationary noise with one-sided output spectrum S_V,

$$C_{\overline{V}}(u) = \int_{f_\ell}^{f_h} S_V(f) \left| \frac{\sin(\pi f T_a)}{\pi f T_a} \right|^2 \cos(2\pi f u)\, \mathrm{d}f. \tag{36}$$

The same averaging window is applied to the transient signal. Squaring Eq. (16) and taking the average gives

$$\sigma_D^2 = (1+\alpha^2) C_{\overline{V}}(0) - 2\alpha C_{\overline{V}}(\Delta t). \tag{37}$$

Substituting Eq. (36) yields Eq. (17), since

$$|1 - \alpha e^{-2\pi i f \Delta t}|^2 = 1 + \alpha^2 - 2\alpha \cos(2\pi f \Delta t). \tag{38}$$

The corrected difference retains a response $1-\alpha$ at zero frequency. The lower integration limit f_ℓ therefore remains part of the noise specification. The calculation uses the continuous average of the output waveform and the circuit noise at its DC operating point. ADC noise and quantisation must be included when a digitisation scheme is chosen.

The noise integral is checked by increasing the spectral sampling density and the quadrature order. The signal is checked by reducing the transient time step. Direct simulations of the adjacent integer ion counts bracket the S/N threshold quoted in the main text.

Figure 8 shows how the averaging time affects the ENC. Increasing the average from 180 µs to 350 µs reduces the ENC from 116.4 to 111.7 electrons RMS. The signal per ion is recalculated for each averaging time.

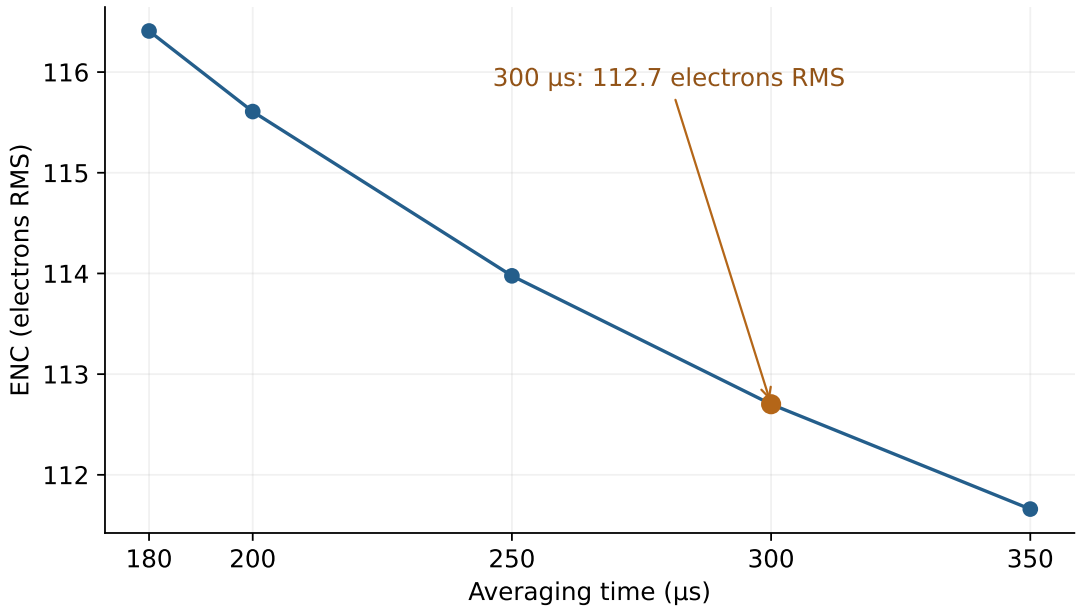


Figure 8: ENC versus averaging time for uniform ion arrivals over 20 ms, with an external pad capacitance of 1.25 pF. The highlighted point is the 300 µs average used in this study.

C Pixel area estimate

The pad is centred in the pixel cell, with a clearance of 50 µm on each side. The electronics are placed outside this region. Each transistor channel is represented by a rectangle with the

dimensions in Table 2; the three parallel instances of M2 are included separately. A margin of 20 µm is added on each side of each channel to allow space for contacts, diffusion, wells and connections. These margins are geometric allowances for the area estimate.

The rectangles are arranged in rows in a strip 200 µm wide, with 20 µm between neighbouring rectangles and 40 µm at the outer boundary. A separate rectangle reserves space for the feedback capacitor and its connections. The resulting CSA block is 840 µm high. The packing calculation checks that the rectangles do not overlap and remain inside the allocated strip. Together with the readout, routing and shared circuitry reserves in Table 5, this gives the approximately 1 mm^2 budget. This is a geometric estimate; a routed layout and parasitic extraction are needed to determine the final circuit area and its effect on the noise.

D Comparison with Topmetal-S

Topmetal-S and NAUSICA use exposed metal electrodes connected to CSAs to measure charge without gas amplification. A focusing electrode guides the ions onto the collecting pads. Topmetal-S was developed to measure both energy and track topology in a high-pressure ion TPC [13]. In ITACA, electroluminescence provides the energy measurement, allowing NAUSICA to choose its pitch and readout for ion-track imaging.

The Topmetal-S collecting electrode is a hexagon of 1 mm diameter, implemented in a 130 nm CMOS process [20]. The NνDEx readout model places these electrodes at 10 mm pitch [21]. NAUSICA uses 0.5 mm square pads at 2 mm pitch, close to the diffusion scale in ITACA. This finer pitch samples the track with more pixels. The smaller pad reduces the capacitive load, while the focusing field provides collection over the pixel cell. The charge measured in each pixel must preserve the track and its blob structure; the precise total-energy measurement is provided by the electron signal.

D.1 Noise measured with charge injection

Topmetal-S reports an ENC of approximately 115 e^- after a digital trapezoidal filter [22]. Repeated injections of a fixed charge produce a distribution of filtered pulse amplitudes. For an injected charge of N_{inj} elementary charges, the ENC is

$$\mathrm{ENC} = N_{\mathrm{inj}} \frac{\sigma_A}{\langle A \rangle}, \tag{39}$$

where $\langle A \rangle$ and σ_A are the mean and standard deviation of that distribution. The quoted noise therefore includes the effect of the filter.

We perform a corresponding injection calculation for the circuit in Table 2, with an external capacitance of 1.25 pF. A pulse carrying 100 e has a 1 µs plateau and 10 ns rise and fall times. We apply a trapezoidal filter with 300 µs rise time and 300 µs flat top. The filtered voltage per elementary charge calibrates the signal. Applying the same filter to the SPICE noise spectrum, including thermal and flicker noise over 0.1 Hz–1 MHz, gives an ENC of 55.4 e^- RMS. The calculation uses the supply and temperature in Table 3 and noise evaluated at the DC operating point.

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	Topmetal-S	NAUSICA
CMOS process	130 nm	180 nm
Collecting electrode	1 mm diameter, hexagonal	0.5 mm side, square
Pitch	10 mm (NνDEx model)	2 mm
Injection ENC	$\sim 115\,e^-$, measured	$55.4\,e^-$, simulated

Table 6: Geometry and charge-injection noise comparison. Both ENC values include trapezoidal filtering. The NAUSICA filter settings are given above; the Topmetal-S report [22] does not specify its trapezoid rise and flat-top durations.

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